

Ema Lovrić, 10.8.2026.

Assignment 16: Responsive web design research

<https://github.com/elovric13/eutech/tree/responsive-web-design-research-elovric13>

1. Research the concept of responsive web design and explain:

- **What responsive design is.**

Responsive web design means a website automatically adjusts its layout depending on the screen size it is being viewed on. The same website looks and behaves differently on a phone, a tablet, and a desktop, but works properly on all of them without the user having to do anything.

- **Why responsive design is important.**

People browse websites on many different devices. If a website only looks good on a desktop but breaks on a phone, a large portion of visitors will have a poor experience and leave. Google also ranks mobile-friendly websites higher in search results, which directly affects how easy it is to find a website online.

- **The benefits for users and businesses.**

For the users, there is no zooming in, no horizontal scrolling, no tiny unreadable text. The experience feels natural regardless of what device the visitor is using.

For businesses, more people can comfortably use the site, which leads to more conversions, lower bounce rates, and better SEO rankings. A single responsive website also eliminates the need to maintain separate desktop and mobile versions.

2. Select three public websites from any industry.

The selected websites are:

- algebra.hr : Sveučiliste Algebra Bernays
- gkzp.hr : Gradska knjižnica Požega
- grbic.hr : Grbić Ltd.

3. Analyze each website using desktop, tablet, and mobile view

- **algebra.hr**

On desktop the layout uses a two-column hero section with a photo on the left and an orange content panel on the right, a full navigation bar with multiple drop-down menus, and a stats row below the fold. On tablet the navigation collapses to a hamburger menu and the layout adapts, though the hero heading gets clipped at the right edge. On mobile the layout becomes fully single-column, the orange panel moves below the image, and the CTA button remains clearly visible.

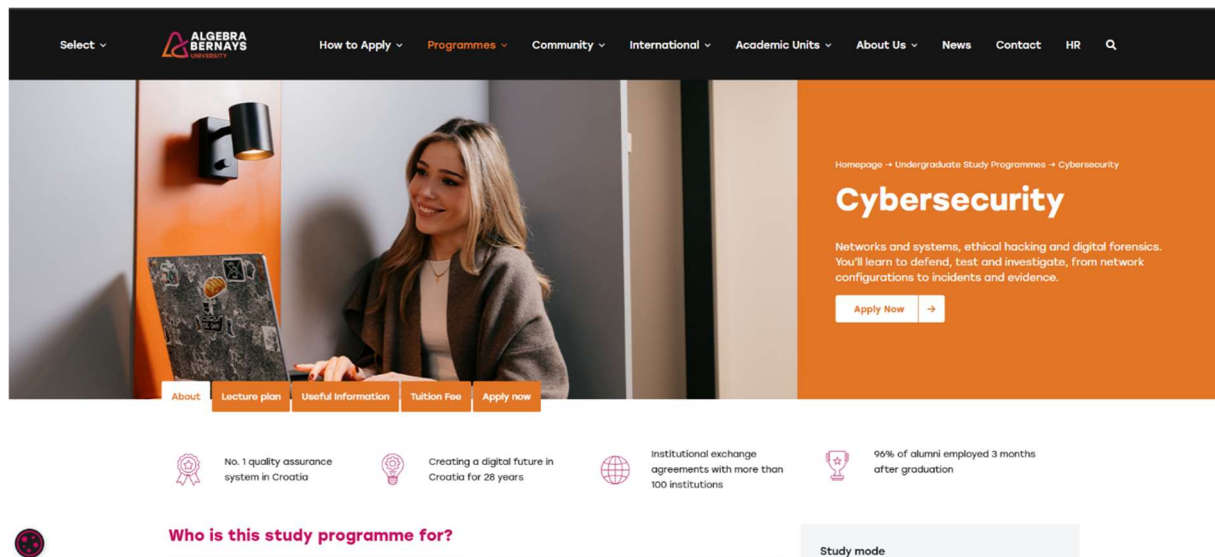


Figure 1: Algebra Bernays - desktop view

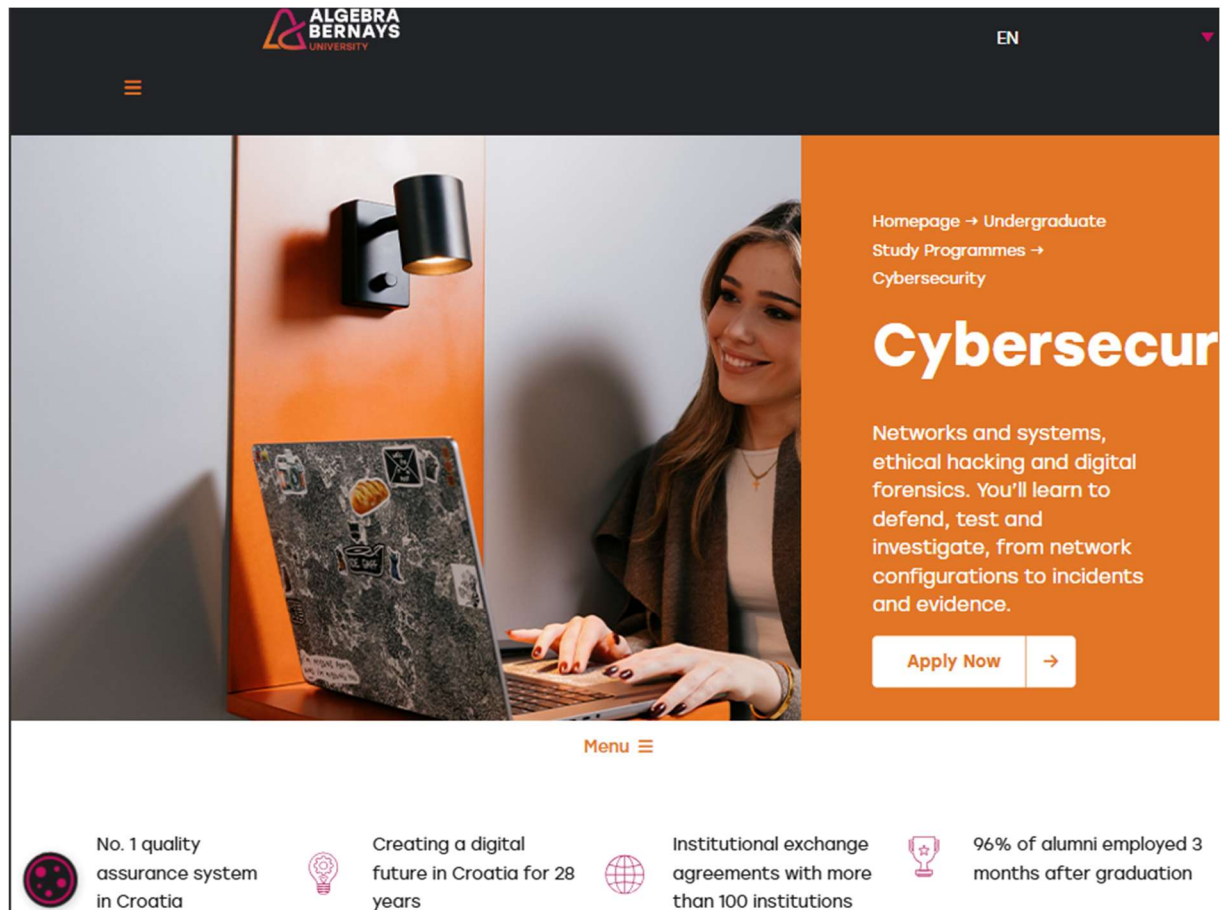


Figure 2: Algebra Bernays - tablet view



Figure 3: Algebra Bernays - mobile view

- **gkpz.hr**

On desktop the site uses a full-width hero image of the library building with overlaid text, a clean navigation bar with drop-downs, and a social media strip at the top. On tablet the navigation collapses to a hamburger and search icon, and the hero image scales down while the text remains overlaid. On mobile the layout becomes single-column and the text wraps naturally, though contrast issues with the hero text become more noticeable on smaller screens.

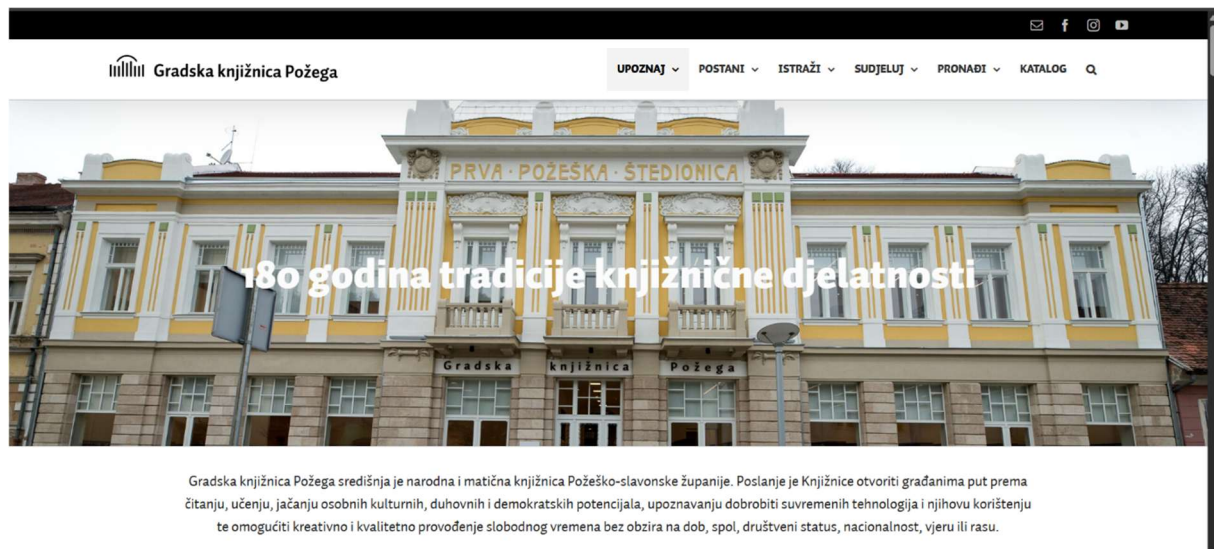


Figure 4: Gradska knjižnica Požega - desktop view



Figure 5: Gradska knjižnica Požega - tablet view



Figure 6: Gradska knjižnica Požega - mobile view

- **grbic.hr**

On desktop the news page uses a two-column layout with the article content on the left and an archive sidebar on the right. The green navigation bar is fully visible with all page links. On tablet the layout begins to compress, the sidebar remains visible but narrower. On mobile the layout becomes single-column, the sidebar disappears, and the navigation collapses to a hamburger menu.

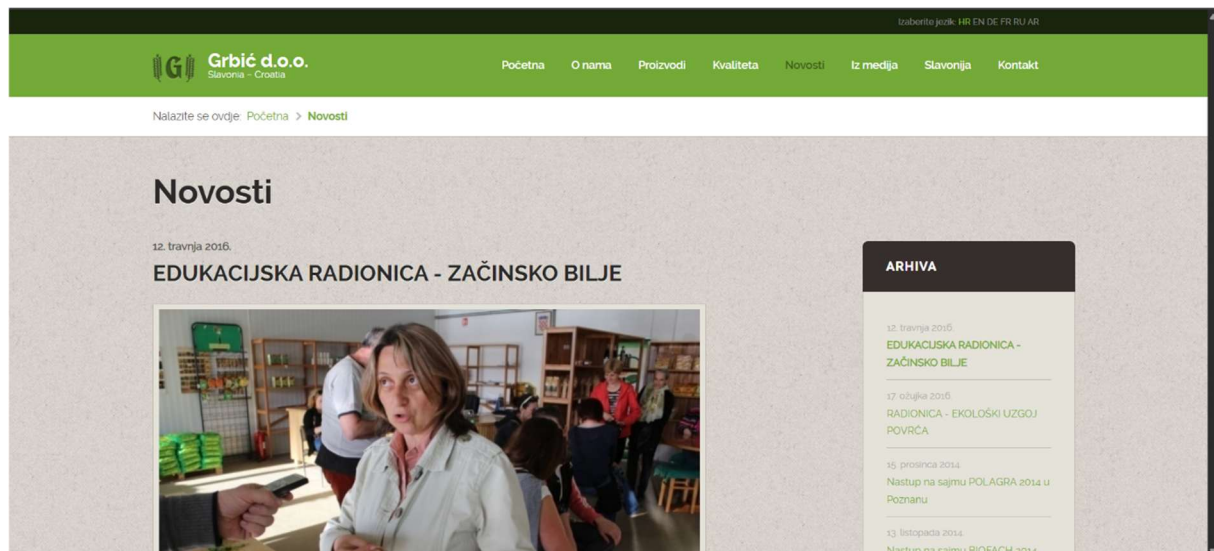


Figure 7: Grbić d.o.o. - desktop view



Figure 8: Grbić d.o.o. - tablet view



Figure 9: Grbić d.o.o. - mobile view

4. Identify responsive design features

- **algebra.hr**

Flexible layout: the two-column hero section stacks vertically on mobile.

Responsive navigation: the full drop-down menu collapses to a hamburger icon on tablet and mobile.

Adaptive images: the hero image scales across all three breakpoints.

Responsive typography: headings and body text resize appropriately on smaller screens.

- **gkpz.hr**

Flexible layout: the single-column structure is maintained across all views.

Responsive navigation: the full navigation with drop-downs collapses to a hamburger and search icon on smaller screens.

Adaptive images: the hero building photo scales well across all three breakpoints.

Responsive typography: heading text wraps and resizes naturally on mobile without breaking the layout.

- **grbic.hr**

Flexible layout: the two-column desktop layout (content and sidebar) collapses to a single column on mobile.

Responsive navigation: the full green navigation bar collapses to a hamburger menu.

Adaptive images: the news article image scales down proportionally on smaller screens.

Responsive typography: headings and body text remain readable across all views.

5. Identify any problems related to responsiveness and explain how they could be improved.

- **algebra.hr**

The heading text overflows and gets clipped on the tablet view. The word Cybersecurity is cut off at the right edge of the screen. This could be fixed by adding a text wrapping or max-width rule for that specific breakpoint.

- **gkpz.hr**

The white italic hero text sits directly over a light-coloured building facade, making it a bit difficult to read. Adding a semi-transparent dark overlay behind the text would ensure readability regardless of the background image at any breakpoint.

- **grbic.hr**

The archive sidebar disappears entirely on mobile, which means users browsing on a phone have no way to navigate between news items without scrolling through the entire page. The sidebar should convert to a collapsible accordion on mobile rather than simply being hidden. For the products part, images are not visible at the mobile view.

6. Summarize the main findings and provide recommendations

All three websites implement the basic principles of responsive design, navigation collapses on smaller screens, images scale, and layouts restructure to single columns on mobile.

Algebra Bernays has the most modern responsive framework and handles the transition between breakpoints well overall, but a specific rendering bug on the tablet view and the disappearing tab navigation are significant issues. Recommendation is fixing the tablet overflow bug.

Gradska knjižnica Požega adapts cleanly across breakpoints and the layout never breaks, but the hero text contrast issue on mobile represents a small readability problem that affects all visitors on small screens. Recommendation is adding dark overlay behind hero text.

Grbić d.o.o. has the most basic responsive implementation of the three. The site functions on mobile but was clearly designed with desktop as the primary view. Recommendation is rebuilding mobile layout from scratch.